

- 7 A solar cell is a device which converts light energy directly into electrical energy. Under the influence of light, a terminal potential difference (p.d.) is generated across the solar cell. This process is explained as follows.

A solar cell consists of three main layers: an N-type material layer, a Junction layer, and a P-type material layer.

There is an electric field across the Junction layer as shown in Fig. 7.1. When photons are incident on the solar cell, electron-hole pairs are generated in the Junction layer. Holes are positively charged carriers with charge $+e$. The electrons and holes move in opposite directions under the influence of the electric field, which sets up a terminal p.d. across the solar cell.

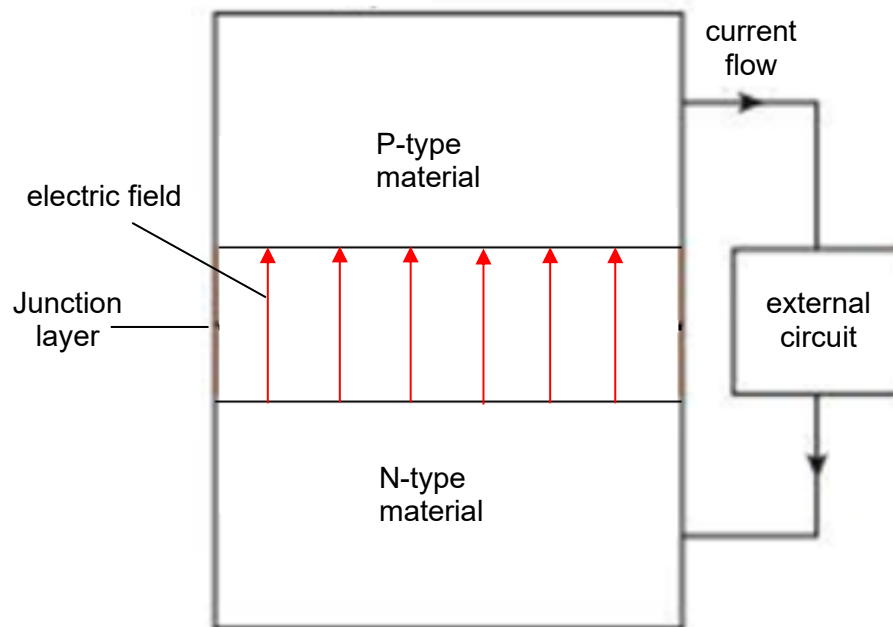


Fig. 7.1

Electron-hole pairs will be generated in the solar cell provided that the incident photon has an energy greater than a certain value known as the band gap energy of the material used.

The magnitude of the p.d. developed across the terminals of the solar cell depends on the intensity of light incident onto the solar cell.

The variation of the potential difference V across the cell with current I can be investigated using the circuit shown in Fig. 7.2

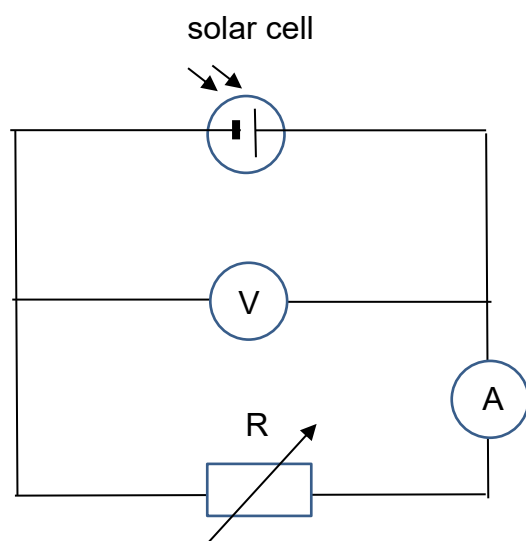


Fig. 7.2

R is a variable resistor with a maximum resistance of $7.0\ \Omega$. The voltmeter has infinite resistance and the resistance of the ammeter is negligible.

Fig 7.3 shows the variation of I with V for a particular cell of surface area $4.0 \times 10^{-4}\ \text{m}^2$ when illuminated normally with light intensity $1100\ \text{W m}^{-2}$.

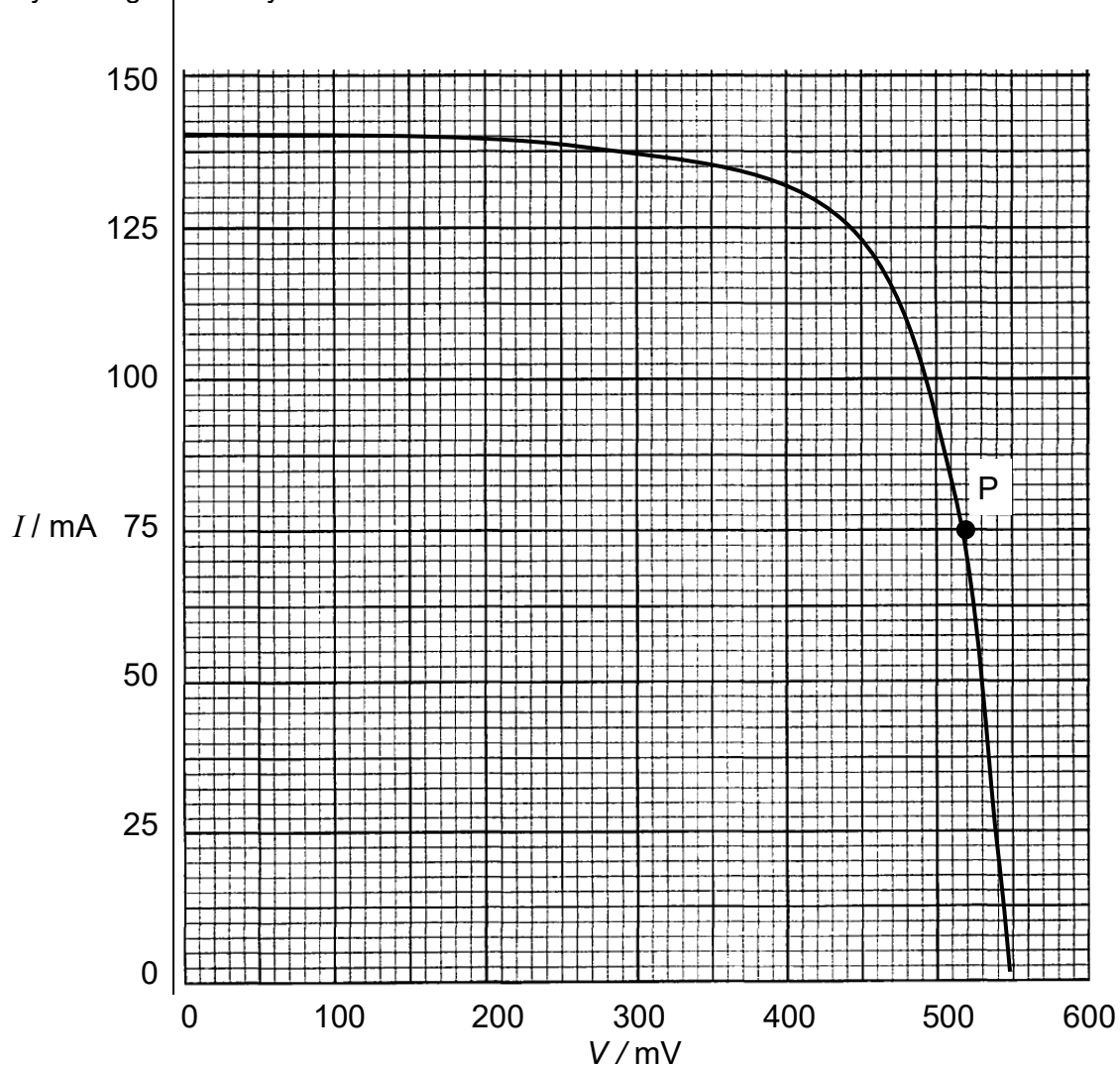


Fig 7.3

- (a) (i) State why a p.d. is set up across the solar cell only for a limited range of frequencies of photons.

.....

 [1]

Solution:

Because the electron-hole pairs are created only when the energy of a photon is greater than the band gap energy of the material of the solar cell, and the energy of a photon is proportional to the frequency of the incident light. 1

1 mark is given to relate the given information (that energy of photon must be greater than the band gap energy of the material) to the knowledge that energy of a photon equals to hf .

- (ii) Explain why the p.d. across the solar cell increases with the intensity of the incident light.

.....

 [2]

Solution:

When intensity is increased at a fixed frequency the number of photons incident per unit time is proportional to the intensity. 1

Since each electron-hole pair created is the result of absorbing one photon, hence more electron-hole pairs will be created per unit time. 1

- (b) (i) Explain how the graph shows that the e.m.f. of the solar cell is 550 mV.

.....

 [1]

Solution:

The value corresponds to an open circuit when the current is zero and the terminal p.d. is equal in value to e.m.f. 1

- (ii) Explain why the experiment data needed to be extrapolated to obtain the trend from P to the point when $V = 550 \text{ mV}$.

.....

 [3]

Solution:

At point P

$I = 75 \text{ mA}$

$V = 520 \text{ mV}$

The resistance at P = $520 \text{ V} / 75 \text{ mA} = 6.9 \Omega$

1

From P to the point when $V = 550 \text{ mV}$, the resistance is larger than 6.9Ω , between 6.9Ω to infinite resistance (as seen from the ratio V/I).

1

However the variable resistor R only has a maximum resistance of 7.0Ω . Thus experiment data cannot be obtained for this range of values of V.

1

- (c) (i) From Fig. 7.3, determine the power dissipation in the load resistor for point P.

power dissipation =W [2]

$I = 75.00 \text{ mA}$

$V = 520 \text{ mV}$

$P = IV = 75 \text{ mA} \times 520 \text{ mV}$

1

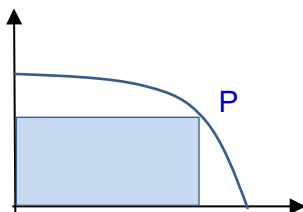
$= 3.90 \times 10^{-2} \text{ mW}$

1

- (ii) On Fig. 7.3, shade an area that represents the power dissipation calculate in c(i).

[1]

Solution:



1

- (iii) From Fig. 7.3, estimate the maximum power dissipation in the load resistor. Show your working.

maximum power =W [2]

When $V = 460 \text{ mV}$ and $I = 120 \text{ mA}$,

$$\text{Power} = 120 \text{ mA} \times 460 \text{ mV} = 55.2 \text{ mW}$$

When $V = 470 \text{ mV}$ and $I = 117.5 \text{ mA}$,

$$\text{Power} = 55.225 \text{ mW}$$

When $V = 480 \text{ mV}$ and $I = 110 \text{ mA}$,

$$\text{Power} = 52.8 \text{ mW}$$

1

Hence Maximum Power $\approx 55 \text{ mW}$

Accept range of Power 55 mW to 60 mW

1

- (iv) On Fig. 7.4 sketch the graph of power output P_o , of the cell against V . Label and include a suitable scale on the vertical axis.

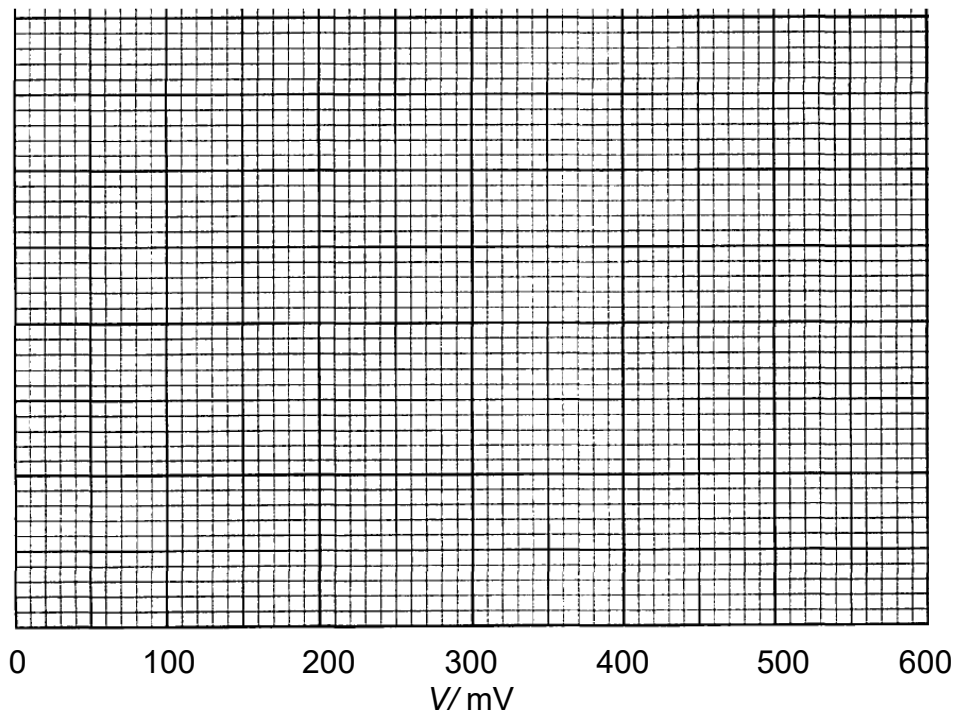
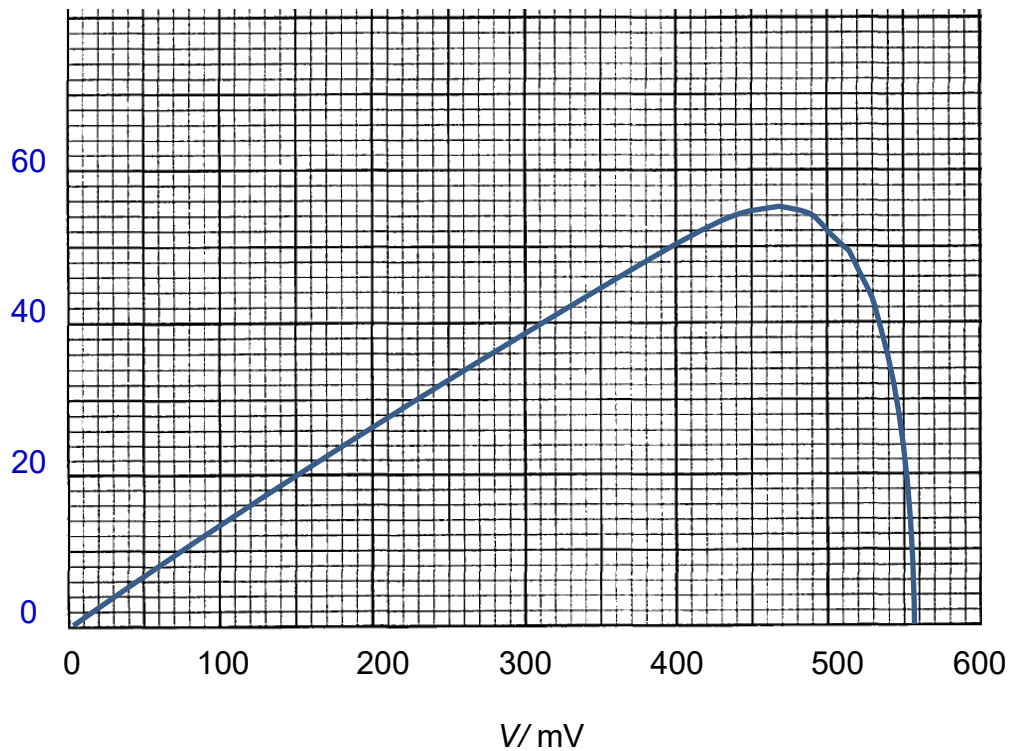


Fig. 7.4

[2]

Solution:

P_0/mW



P_0 increases and decreases

$P_0 = 0$ at $V = 0$, and V is max at 550 mV, and $P_0 = 55$ mW at $V = 470$ mV
(estimated from area of 'rectangle' under the I-V graph)

1

P_0 increases almost linearly for $V < 200$ mV

(because $P_0 = IV$, so when I is approximately constant, P_0 is approximately proportional to $V \rightarrow$ linear variation)

1

(d) Calculate the maximum efficiency of conversion of light energy into electrical energy.

maximum efficiency = [2]

From (c)(iii), Maximum power output = 55 mW

Power input (from incident light energy) = intensity $\times$ area = $1100 \times 4 \times 10^{-4} = 440$ mW

1

efficiency = $55 / 440 = 0.125$ or 12.5 %

1

- (e) A number of solar cells are connected to a load resistor L as shown in Fig. 7.5.

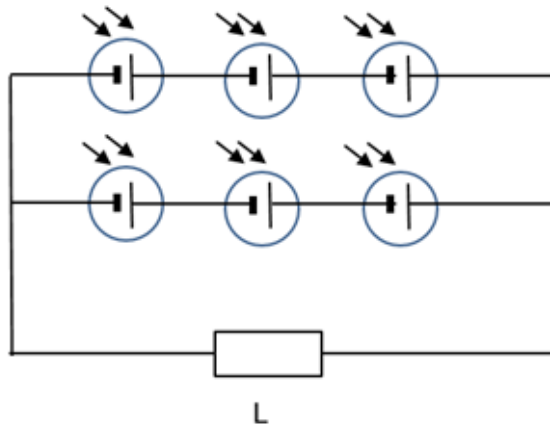


Fig. 7.5

The resistance of L has been adjusted so that each cell gives the maximum power estimated in (c)(iii).

Calculate

- (i) the potential difference across L,

potential difference =V [1]

Solution:

$$V = 3 \times 470 \text{ mV} = 1.41 \text{ V}$$

1

- (ii) the current through L.

current =A [1]

Solution:

$$I = 2 \times 117.5 \text{ mA} = 0.235 \text{ A}$$

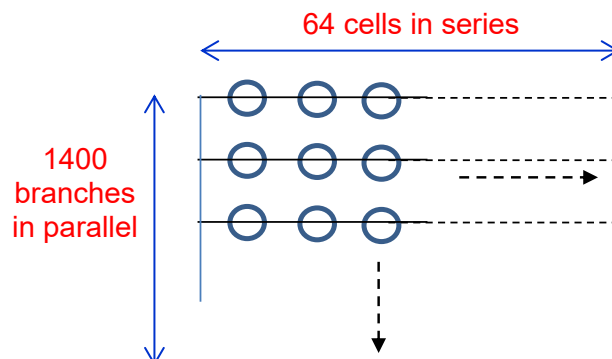
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- (f) Suppose each cell is operating at maximum power estimated in (c)(iii). Draw a suitable network of cells so that the cells may be used to provide an output power of approximately 5 kW at 30 V.

[3]

Solution:

A network of cells in series and parallel connections



1 Mark – Correct number of cells in series

1 Mark – Correct number of cells in parallel

1 Mark – Diagram - Label the number of cells in series per branch and number of branches (Circuit symbol in branches not required)

At max power, each cell has a p.d. of 47 mV and current of 117.5 mA

Number of cells **in series** = $30 / 0.470 = 63.8 = 64$ cells
Number of cells **required** = $5 \text{ kW} / 55 \text{ mW} = 90900$ cells
Number of **parallel circuits** = $90900 / 64 = 1400$

-- END OF PAPER 2 --